Ensemble Bayesian max-margin one-class classifier for SAR target discrimination in complex scenes

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Abstract

Synthetic aperture radar (SAR) target discrimination is the key stage in a typical SAR automatic target recognition system, and directly influences the performance of the whole system. However, the performance of some traditional discriminators is degraded in some complex scenes and sensitive to the hyperparameters. To enhance discrimination performance in complex scenes and reduce the sensitivity to the model parameters, a novel discrimination method, ensemble Bayesian max-margin one-class classifier (EnB-MMOCC), is presented in the paper. In EnB-MMOCC, Dirichlet process mixture is introduced to cluster the input data, and a max-margin one-class classifier is learned with max-margin criteria in each cluster. With the ensemble of some simple classifiers, the complex nonlinear classification can be implemented to enhance the classification performance. Experimental results based on real SAR data demonstrate that EnB-MMOCC can not only enhance discrimination performance, but also be robust to the model parameters.

1 Introduction

Automatic target recognition (ATR) based on synthetic aperture radar (SAR) images is a widely used technique in the military and civilian fields [1-3]. The typical SAR ATR system developed by the Lincoln Laboratory consists three stages [1]: detection, discrimination, and classification/recognition. Since target discrimination is the key stage linking detection and classification/recognition, it has attracted much attention during the last two decades [4-10]. For examples, Wang et al. proposed a target discrimination method based on visual attention [4]. In [4], modified saliency and gist (MSG) features are extracted firstly, and then support vector machine (SVM) is used as the discriminator. A superpixel-level target discrimination is proposed in [5], which extracts multi-level and multi-domain feature descriptor are extracted based on the superpixel-level target detection results, and then SVM is used as the discriminator. Li et al. proposed a target discrimination method, in which scattering center feature are extracted from SAR images, then K-center One-Class Classifier is used to classify target and clutter [6]. Park et al. proposed a target discrimination method based on size-related features quadratic distance (QD) classifier [7]. Gao proposed an improved scheme for target discrimination in High-Resolution SAR Images [8]. In [8], spatial boundary features, fractal features and contrast feature are extracted firstly, and then Genetic algorithm is used for feature selection. Finally, weighted QD is utilized as the discriminator.

Although various discrimination methods are proposed, most of them focus on feature extraction and ignore the design of discriminator. Consequently, the performance of these methods is degraded in some complex cases and sensitive to the hyperparameters of the discriminator, such as the value of k in k-center or k-means algorithms, and the kernel parameter in SVM or one-class SVM. To enhance the discrimination performance in the complex scenes and robustness to the model parameter, a novel target discriminator, namely, ensemble Bayesian max-margin one-class classifier (EnB-MMOCC), is proposed for target discrimination. In EnB-MMOCC, Dirichlet process mixture (DPM) is used to cluster the training data, and a max-margin one-class classifier is learned in each cluster via max-margin criteria. With the ensemble of some simple classifiers, the complex nonlinear discrimination boundary can be constructed to enhance the classification performance.

The main contributions of our method are summarized in the following:

- In EnB-MMOCC, DPM and max-margin one-class classifier are joint optimized to implement the supervised clustering.
- In max-margin one-class classifier, a novel feature transformation with explicit expression is proposed to modify the one-class SVM.
- In EnB-MMOCC, the complex nonlinear discrimination boundary can be constructed with the ensemble of several simpler one-class classifier to improve the discrimination accuracy in the complex scenes.

The remainder of the paper is organized as follows. The proposed method is presented in Section 2. The performance of our method is evaluated on real synthetic aperture radar (SAR) data in Section 3. A conclusion about our method is given in Section 4.

2. Ensemble Bayesian max-margin one-class classifier

2.1 Bayesian max-margin one-class classifier

In max-margin one-class classifier, a novel feature transformation with explicit expression is proposed to modify the one-class SVM [9]. Moreover, the model parameters of max-margin one-class classifier are obtained with Bayesian solution. The objective function of max-margin one-class classifier is shown in Eq. (1):

$$\begin{aligned}&\min_{\mathbf{w},\xi_i} \quad \frac{1}{2}\|\mathbf{w}\|^2+\frac{1}{\eta N}\sum_i \xi_i-1\\ &s.t. \quad \left(\mathbf{w}^T\tilde{\mathbf{x}}_i\right)\geq 1-\xi_i,\forall i \\ &\qquad\quad \xi_i\geq 0,\forall i\end{aligned} \tag{1}$$

where $\mathbf{w}$ is the slope of the classification hyperplane, ξ_i is the slack variable, $\tilde{\mathbf{x}}_i$ is the transformation of $\mathbf{x}_i$ based on the following formulation:

$$\tilde{\mathbf{x}}_i=\left[\kappa\left(\mathbf{x}_i,\mathbf{r}_1\right),\kappa\left(\mathbf{x}_i,\mathbf{r}_2\right),\ldots,\kappa\left(\mathbf{x}_i,\mathbf{r}_M\right)\right] \tag{2}$$

where Gaussian kernel function $\kappa(\mathbf{a}_1,\mathbf{a}_2)=\exp(-\|\mathbf{a}_1-\mathbf{a}_2\|_2^2/2\sigma^2)$ with hyperparameter σ, $\mathbf{x}_i$ is the original training sample, and $\mathbf{r}=\{\mathbf{r}_1,\mathbf{r}_2,\ldots,\mathbf{r}_M\}$ is the base vector set by selecting a certain number of training samples. Gaussian kernel function has the characteristics of similarity preserving and the value of $\kappa\left(\mathbf{x}_i,\mathbf{r}_j\right)$ is $0<\kappa\left(\mathbf{x}_i,\mathbf{r}_j\right)\leq 1$, therefore, the transformed data are located in the unit hypersquare in the first quadrant. Specifically, the clutter samples are scattered near the origin and the target samples are scattered away from the origin.

In general, the model parameters in Eq. (1) can be solved with convex optimization algorithms, such as sequential minimal optimization (SMO) algorithm [10]. However, in order to achieve the joint optimization of modified OCSVM and DPM, we propose a Bayesian solution for Eq. (1).

Firstly, Eq. (1) can be rewritten as Eq. (3):

$$\min_{\mathbf{w}} \; d\left(\mathbf{w}\right)=\frac{1}{2}\|\mathbf{w}\|^2+2l\sum_i \max\left(1-\mathbf{w}^T\tilde{\mathbf{x}}_i,0\right) \tag{3}$$

Based on [11], the optimal parameter $\mathbf{w}_{opt}$ is obtained based on Eq. (4):

$$\mathbf{w}_{opt}=\int p(\mathbf{w}\mid\tilde{\mathbf{x}})d\mathbf{w} \tag{4}$$

where the formulation of $p(\mathbf{w}\mid\tilde{\mathbf{x}})$ is shown in Eq. (5):

$$\begin{aligned}&p(\mathbf{w}\mid\tilde{\mathbf{x}})\propto\exp\left(-d\left(\mathbf{w}\right)\right)\propto p(\tilde{\mathbf{x}}\mid\mathbf{w})p(\mathbf{w})\\ &p\left(\mathbf{w}\right)=\exp\left(-\frac{1}{2}\mathbf{w}^T\mathbf{w}\right)\\ &p\left(\tilde{\mathbf{x}}\mid\mathbf{w}\right)=\exp\left(-2l\sum_i\max\left(1-\mathbf{w}^T\tilde{\mathbf{x}}_i,0\right)\right)\end{aligned} \tag{5}$$

Based on the data augmentation technique [12], latent variable λ_i is introduced to rewritten $p\left(\tilde{\mathbf{x}}\mid\mathbf{w}\right)$ as Eq. (6):

$$\begin{aligned}p_i\left(\tilde{\mathbf{x}}_i\mid\mathbf{w}\right)&=\exp\left(-2l\max\left(1-\mathbf{w}^T\tilde{\mathbf{x}}_i,0\right)\right)\\ &=\int_0^\infty\frac{1}{\sqrt{2\pi\lambda_i}}\exp\left\{\frac{\left(l-l\mathbf{w}^T\tilde{\mathbf{x}}_i+\lambda_i\right)^2}{2\lambda_i}\right\}d\lambda_i\end{aligned} \tag{6}$$

Therefore, the distribution $p\left(\mathbf{w},\lambda\mid\tilde{\mathbf{x}}\right)$ can be obtained as Eq. (7):

$$\begin{aligned}&p\left(\mathbf{w},\lambda\mid\tilde{\mathbf{x}}\right)\propto p\left(\tilde{\mathbf{x}},\lambda\mid\mathbf{w}\right)p\left(\mathbf{w}\right)\\ &\propto\prod_i\frac{1}{\sqrt{2\pi\lambda_i}}\exp\left\{\frac{\left(l-l\mathbf{w}^T\tilde{\mathbf{x}}_i+\lambda_i\right)^2}{2\lambda_i}\right\}\times\exp\left(-\frac{1}{2}\mathbf{w}^T\mathbf{w}\right)\end{aligned} \tag{7}$$

Based on Markov Chain Monte Carlo (MCMC) algorithm, parameter reference in Eq. (7) can be performed using Gibbs sampling.

2.2 Ensemble Bayesian max-margin one-class classifier

EnB-MMOCC is the ensemble of several max-margin one-class classifiers based on DPM clustering. In EnB-MMOCC, DPM is used to cluster the training data, and a max-margin one-class classifier is learned in each cluster via max-margin criteria. Specially, In EnB-MMOCC, DPM and max-margin one-class classifier are joint optimized to implement the supervised clustering.

In EnB-MMOCC, the representation of DPM is the stick-breaking model [13]. In detail, the distribution of data in each cluster is Gaussian distribution $N(\mathbf{x}_i;\mathbf{u}_{z_i},\mathbf{\Sigma}_{z_i})$, and the base distribution is a conjugate distribution of Gaussian distribution, i.e., Normal-Wishart distribution $\{\mathbf{u}_c,\mathbf{\Sigma}_c\}_{c=1}^C\sim NW\left(\mathbf{u}_c,\mathbf{\Sigma}_c;\mathbf{u}_0,\mathbf{\Sigma}_0,\beta_0,\gamma_0\right)$. Therefore, EnB-MMOCC model can be written as:

$$\begin{aligned}&v_c\sim Beta\left(v_c;1,\alpha\right),\{\mathbf{u}_c,\mathbf{\Sigma}_c\}_{c=1}^C\sim NW\left(\mathbf{u}_c,\mathbf{\Sigma}_c;\mathbf{u}_0,\mathbf{\Sigma}_0,\beta_0,\gamma_0\right)\\ &\pi_c(\mathbf{v})=v_c\prod_{j=1}^{c-1}\left(1-v_j\right),z_i\sim Multi(z_i;\boldsymbol{\pi})\\ &\mathbf{x}_i\mid z_i,\{\mathbf{u}_c,\mathbf{\Sigma}_c\}_{c=1}^C\sim N(\mathbf{x}_i;\mathbf{u}_{z_i},\mathbf{\Sigma}_{z_i})\end{aligned} \tag{8}$$

$$\begin{aligned}&\{\mathbf{w}_c\}_{c=1}^C\sim N\left(\mathbf{w}_c;0,\mathbf{I}\right)\\ &\tilde{\mathbf{x}}_i,\lambda_i\mid z_i,\{\mathbf{w}_c\}_{c=1}^C\propto\frac{1}{\sqrt{2\pi\lambda_i}}\exp\left\{\frac{\left(l-l\mathbf{w}_{z_i}{}^T\tilde{\mathbf{x}}_i+\lambda_i\right)^2}{2\lambda_i}\right\}\end{aligned} \tag{9}$$

Eq. (8) is DPM model with the stick-breaking representation, and Eq. (9) is the model of max-margin one-class classifier. Based on Eq. (8) and Eq. (9), the distribution $p\left(\{\mathbf{w}_c\}_{c=1}^C,\{\mathbf{u}_c,\mathbf{\Sigma}_c\}_{c=1}^C,\mathbf{v},\boldsymbol{\lambda},\mathbf{z}\mid\mathbf{x},\tilde{\mathbf{x}}\right)$ can be derived as follows:

$$\begin{aligned}&p\left(\{\mathbf{w}_c\}_{c=1}^C,\{\mathbf{u}_c,\mathbf{\Sigma}_c\}_{c=1}^C,\mathbf{v},\boldsymbol{\lambda},\mathbf{z}\mid\mathbf{x},\tilde{\mathbf{x}}\right)\\ &\propto p\left(\{\mathbf{w}_c\}_{c=1}^C,\{\mathbf{u}_c,\mathbf{\Sigma}_c\}_{c=1}^C,\mathbf{v},\boldsymbol{\lambda},\mathbf{z},\mathbf{x},\tilde{\mathbf{x}}\right)\\ &=p\left(\mathbf{x}\mid\{\mathbf{u}_c,\mathbf{\Sigma}_c\}_{c=1}^C,\mathbf{z}\right)\prod_{c=1}^C p\left(\mathbf{u}_c,\mathbf{\Sigma}_c\mid\mathbf{u}_0,\mathbf{\Sigma}_0,\beta_0,\gamma_0\right)\\ &\cdot p\left(\mathbf{z}\mid\mathbf{v}\right)p\left(\mathbf{v}\mid\alpha_0\right)\prod_{c=1}^C p\left(\tilde{\mathbf{x}},\boldsymbol{\lambda}\mid\mathbf{w}_c\right)p\left(\mathbf{w}_c\right)\end{aligned} \tag{10}$$

where $\mathbf{z}=\{z_i\}_{i=1}^N$, $\boldsymbol{\lambda}=\{\lambda_i\}_{i=1}^N$. The conditional posterior of each model parameter can be derived via Eq. (10). Based on the conditional posteriors of each model parameter, the model parameters can be approximated with the MCMC method and Gibbs sampler as shown in [14].

Fig. 1 is the Graphical model representation of EnB-MMOCC, where red dotted box is the DPM clustering model, and the blue dotted box is the max-margin one-class classifier model. We can see from Fig. 1 that DPM clustering and max-margin one-class classifier link up with each other via the variable of clustering index z_i.

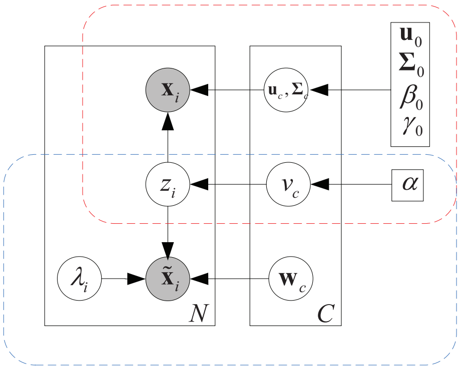


Figure 1 Graphical model representation of EnB-MMOCC

3 Experimental results

The real SAR data, namely Sandia Manias dataset, is used to evaluate the effectiveness of EnB-MMOCC in this Section. Nine SAR images are selected from this dataset, and Fig. 2 is an example of SAR image used in the experiments. In this dataset, the target class is the car, and the clutter class includes grassland, building, trees and road.

Figure 2 An example of SAR image in SAR real dataset.

Based on the constant false alarm rate detection algorithm, 260 target patches and 86 clutter patches are obtained from nine SAR images. Fig. 3 exhibits some patch examples, where the first row is the target patches, and the second row is the clutter patches.

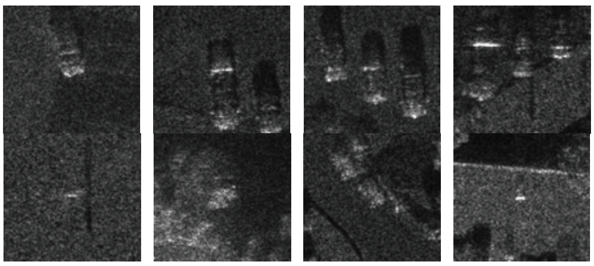


Figure 3 Some target and clutter examples of patch from nine SAR images.

Three criterions are introduced to evaluate one-class classifier: pe, i.e., discrimination accuracy, F1-Score, and Area Under the Receiver Operating Characteristic Curve (AUC).

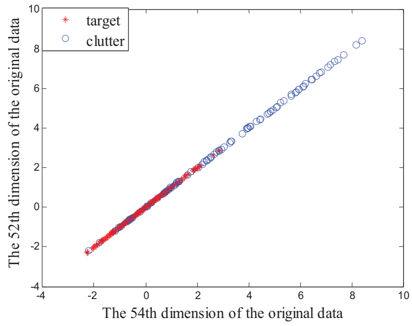
In the experiments, we randomly select 150 target samples as the training samples, and the testing dataset includes the remaining target samples and all clutter samples. We conduct 20 random experiments to remove the randomness of the results. Table 1 gives the results of our method and some traditional discriminators.

Table 1 The Results (%) of Our Method and Some Traditional Discriminators

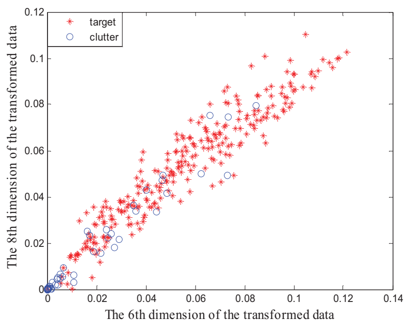
Method	pe	F1-Score	AUC
L-SVDD [17]	82.14	85.05	87.30
One-class SVM [9]	82.97	87.86	85.27
k -means [16]	84.82	86.97	88.46
PCA [15]	73.83	79.11	77.99
MST [18]	84.13	87.05	89.08
SOM [19]	80.23	82.31	89.05
Auto-encoder [16]	83.42	85.98	87.65
MPM [20]	84.67	86.85	89.08
LPDD [21]	83.24	85.11	88.36
EnB-MMOCC	***85.43***	***88.31***	***89.33***

We can see from Table 1 that EnB-MMOCC obtains the highest value of discrimination accuracy, F1-Score, and AUC. Therefore, EnB-MMOCC can really improve the discrimination performance in the complex scenes compared with traditional one-class classifiers.

To verify the effectiveness of the proposed transformation, the 2-D feature visualization of the original and the transformed features are shown in Figs. 4 (a)-(b) respectively. Fisher criterion is utilized to selected two features with the top two scores. The 2-D feature visualization of the original features is shown in Fig. 4 (a), while Fig. 4 (b) is the 2-D feature visualization of the transformed features.



a



b

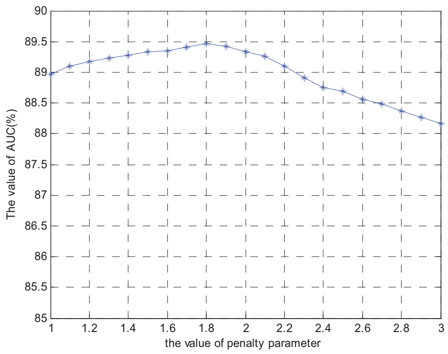
Figure 4 The 2-D Feature Visualization of Original and Transformed Features.

(a) 21th versus 53th for original features

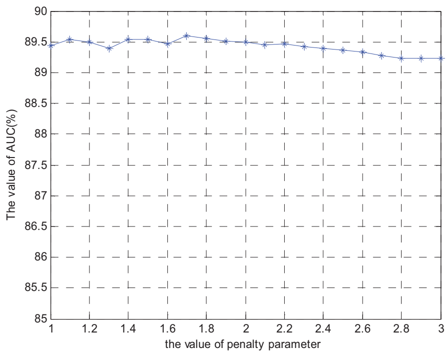
(b) 6th versus 8th for transformed features

It is apparent that the original target and clutter samples mix with each other. Specifically, most of transformed clutter data is located near the origin, and the corresponding target data is located away from the origin. Therefore, our feature transformation is effective for one-class SVM.

Finally, we analyze the robustness of EnB-MMOCC and one-class SVM to the model hyperparameter, i.e., Gaussian kernel parameter. Fig. 5 (a) shows the AUCs of one-class SVM with the value of hyperparameter σ, while Fig. 5 (b) gives the AUCs of EnB-MMOCC with the value of hyperparameter σ. The range of Gaussian kernel parameter is $\left[\sigma_{opt}-1,\sigma_{opt}+1\right]$, where σ_{opt} is the value corresponding to the optimal testing results. We can see that the AUC of our method is stable for different values of hyperparameter σ while the AUC of one-class is sensitive to the value of hyperparameter σ.



a



b

Figure 5 The AUC of KOCSVM and EnB-MMOCC with different values of the Gaussian kernel parameter.
(a) KOCSVM
(b) EnB-MMOCC

4 Conclusion

A novel one-class classifier, namely, EnB-MMOCC, is proposed to enhance the discrimination performance in the complex scenes and improve the robustness to the model parameter. In EnB-MMOCC, DPM is used to cluster the training data, and a max-margin one-class classifier is learned in each cluster via max-margin criteria. Specifically, DPM and max-margin one-class classifier are joint optimized to implement the supervised clustering. With the ensemble of some simple classifiers, the complex nonlinear discrimination boundary can be constructed to enhance the classification performance. Moreover, in max-margin one-class classifier, a novel feature transformation with explicit expression is proposed to modify the one-class SVM. Experiments on the MiniSAR dataset demonstrate the effectiveness of EnB-MMOCC.

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6 References

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